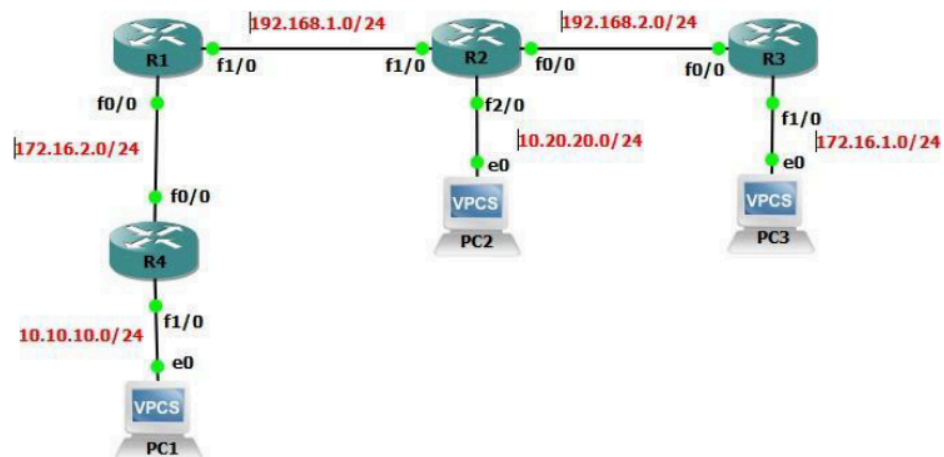
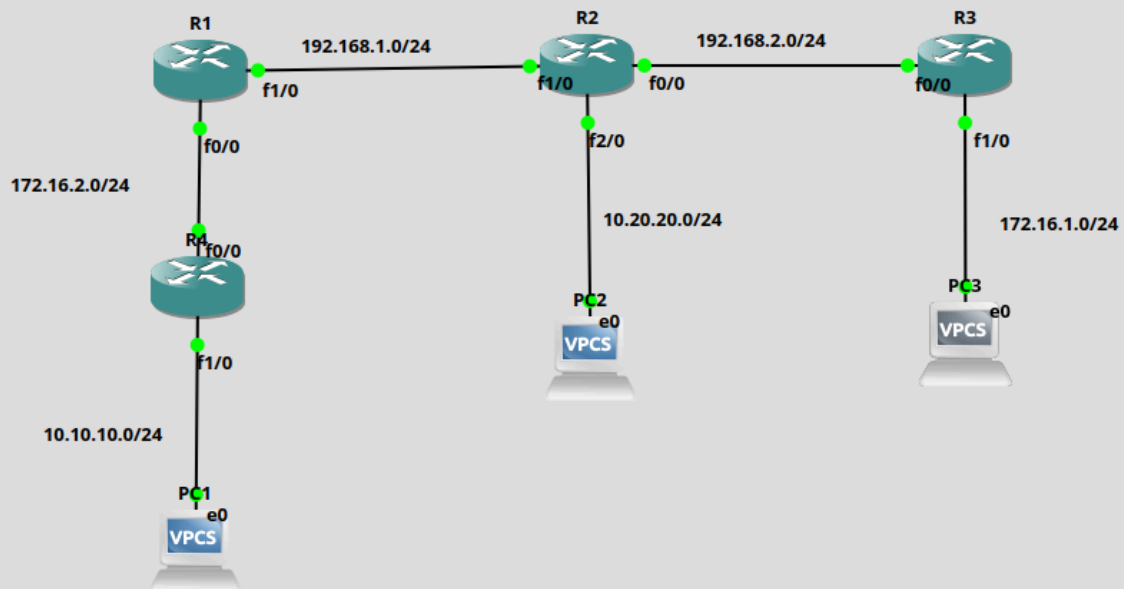


## TP1 - R302

### TP1: Border Gateway Protocol (BGP)



Reproduire le schéma avec les routeurs:



## Configuration IP Routeur 1 :

```
en
!  
conf t  
!  
interface FastEthernet0/0  
ip address 172.16.2.1 255.255.255.0  
no sh  
duplex auto  
speed auto  
!  
interface FastEthernet1/0  
ip address 192.168.1.1 255.255.255.0  
no sh  
duplex auto  
speed auto  
!
```

## Configuration IP routeur 2

```
en
!  
conf t  
!  
interface FastEthernet0/0  
ip address 192.168.2.2 255.255.255.0  
no sh  
duplex auto  
speed auto
```

```
!  
interface FastEthernet1/0  
 ip address 192.168.1.2 255.255.255.0  
 no sh  
 duplex auto  
 speed auto
```

```
!  
interface FastEthernet2/0  
 ip address 10.20.20.2 255.255.255.0  
 no sh  
 duplex auto  
 speed auto
```

```
!  
Configuration IP Routeur 3 :
```

```
en  
!  
conf t  
!  
interface FastEthernet0/0  
 ip address 192.168.2.3 255.255.255.0  
 no sh  
 duplex auto  
 speed auto
```

```
!  
interface FastEthernet1/0  
 ip address 172.16.1.3 255.255.255.0  
 no sh  
 duplex auto  
 speed auto
```

```
!
```

## Configuration IP routeur 4

```
en
```

```
!
```

```
conf t
```

```
!
```

```
interface FastEthernet0/0
```

```
ip address 172.16.2.4 255.255.255.0
```

```
no sh
```

```
duplex auto
```

```
speed auto
```

```
!
```

```
interface FastEthernet1/0
```

```
ip address 10.10.10.4 255.255.255.0
```

```
no sh
```

```
duplex auto
```

```
speed auto
```

```
!
```

## Configuration pour les @IP

Configuration IP PC1 :

```
ip 10.10.10.5/24 10.10.10.4
```

Configuration IP PC2 :

```
ip 10.20.20.5/24 10.20.20.2
```

Configuration IP PC3 :

```
ip 172.16.1.5/24 172.16.1.3
```

**Configurer les routeurs avec un routage statique.  
Fournir la configuration complète de chaque routeur  
ainsi que les tables de routage (utilisez show ip route  
pour vérifier les routes apprises).**

```
R4#show ip route
```

```
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP  
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area  
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2  
E1 - OSPF external type 1, E2 - OSPF external type 2  
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2  
ia - IS-IS inter area, * - candidate default, U - per-user static route  
o - ODR, P - periodic downloaded static route
```

```
Gateway of last resort is not set
```

```
172.16.0.0/24 is subnetted, 2 subnets  
B    172.16.1.0 [20/0] via 172.16.2.1, 00:10:05  
C    172.16.2.0 is directly connected, FastEthernet0/0  
10.0.0.0/24 is subnetted, 2 subnets  
B    10.20.20.0 [20/0] via 172.16.2.1, 00:10:05  
C    10.10.10.0 is directly connected, FastEthernet1/0  
B    192.168.1.0/24 [20/0] via 172.16.2.1, 00:10:36  
B    192.168.2.0/24 [20/0] via 172.16.2.1, 00:10:05  
R4#
```

```
R3(config)#do show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route
```

Gateway of last resort is not set

```
       172.16.0.0/24 is subnetted, 2 subnets
C       172.16.1.0 is directly connected, FastEthernet1/0
B       172.16.2.0 [20/0] via 192.168.2.2, 00:10:35
       10.0.0.0/24 is subnetted, 2 subnets
B       10.20.20.0 [20/0] via 192.168.2.2, 00:10:35
B       10.10.10.0 [20/0] via 192.168.2.2, 00:10:35
B       192.168.1.0/24 [20/0] via 192.168.2.2, 00:10:35
C       192.168.2.0/24 is directly connected, FastEthernet0/0
R3(config)#
```

```
R2(config)#do sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route
```

Gateway of last resort is not set

```
       172.16.0.0/24 is subnetted, 2 subnets
B       172.16.1.0 [20/0] via 192.168.2.3, 00:11:33
B       172.16.2.0 [20/0] via 192.168.1.1, 00:11:07
       10.0.0.0/24 is subnetted, 2 subnets
C       10.20.20.0 is directly connected, FastEthernet2/0
B       10.10.10.0 [20/0] via 192.168.1.1, 00:11:07
C       192.168.1.0/24 is directly connected, FastEthernet1/0
C       192.168.2.0/24 is directly connected, FastEthernet0/0
R2(config)#
```

```

R1#sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    172.16.0.0/24 is subnetted, 2 subnets
B       172.16.1.0 [20/0] via 192.168.1.2, 00:11:42
C       172.16.2.0 is directly connected, FastEthernet0/0
    10.0.0.0/24 is subnetted, 2 subnets
B       10.20.20.0 [20/0] via 192.168.1.2, 00:12:12
B       10.10.10.0 [20/0] via 172.16.2.4, 00:12:13
C       192.168.1.0/24 is directly connected, FastEthernet1/0
B       192.168.2.0/24 [20/0] via 192.168.1.2, 00:12:12
R1#

```

**Configurer maintenant le réseau en utilisant eBGP avec les systèmes autonomes suivants :**

- **R4 dans l'AS 65100**
- **R1 dans l'AS 65101**
- **R2 dans l'AS 65102**
- **R3 dans l'AS 65103**

**Configuration BGP Routeur 1 :**

**router bgp 65101**

**!**

```
neighbor 192.168.1.2 remote-as 65102
!  
neighbor 172.16.2.4 remote-as 65100
!  
network 192.168.1.0 mask 255.255.255.0
!  
network 172.16.2.0 mask 255.255.255.0
!
```

Configuration BGP Routeur 2:

```
router bgp 65102
!  
neighbor 192.168.1.1 remote-as 65101
!  
neighbor 192.168.2.3 remote-as 65103
!  
network 192.168.1.0 mask 255.255.255.0
!  
network 192.168.2.0 mask 255.255.255.0
!  
network 10.20.20.0 mask 255.255.255.0
!
```

Configuration BGP Routeur 3;

```
router bgp 65103
!  
neighbor 192.168.2.2 remote-as 65102
```



```
!
network 192.168.2.0 mask 255.255.255.0
```

```
!
network 172.16.1.0 mask 255.255.255.0
!
```

Configuration BGP Routeur 4;

```
router bgp 65100
```

```
!
neighbor 172.16.2.1 remote-as 65101
```

```
!
network 172.16.2.0 mask 255.255.255.0
```

```
!
network 10.10.10.0 mask 255.255.255.0
!
```

```
Success rate is 100 percent (5/5), round-trip min/avg/max = 4/11/24 ms
R2#ping 10.20.20.5
```

```
Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 10.20.20.5, timeout is 2 seconds:
```

```
.!!!!
```

```
Success rate is 80 percent (4/5), round-trip min/avg/max = 4/11/24 ms
```

```
R2#ping 192.168.2.3
```

```
Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 192.168.2.3, timeout is 2 seconds:
```

```
!!!!!
```

```
Success rate is 100 percent (5/5), round-trip min/avg/max = 8/10/16 ms
```

```
R2#
```

Même question en eBGP entre R3 & R2 et en iBGP entre R1-R2-R4. Est ce que cela fonctionne ?

Apporter les corrections nécessaires au schéma et configuration

## Configuration iBGP :

On configure le iBGP sur R1, R2 et R4.

**Configuration iBGP sur R1 :**

```
router bgp 65101
```

```
!
```

```
no neighbor 192.168.1.2 remote-as 65102
```

```
!
```

```
no neighbor 172.16.2.4 remote-as 65100
```

```
!
```

```
network 192.168.1.0 mask 255.255.255.0
```

```
!
```

```
network 172.16.2.0 mask 255.255.255.0
```

```
!
```

**Configuration iBGP Routeur 2**

```
no router bgp 65102
```

```
!
```

```
router bgp 65101
```

```
!
```

```
no neighbor 192.168.1.1 remote-as 65101
```

```
!
```

```
neighbor 192.168.2.3 remote-as 65103
```

```
!
```

```
network 192.168.1.0 mask 255.255.255.0
```

```
!
```

```
network 192.168.2.0 mask 255.255.255.0
```

```
!
```

```
network 10.20.20.0 mask 255.255.255.0
```

```
!
```

```
network 10.30.30.0 mask 255.255.255.0
```

```
!
```

```
end
```

```
!
```

```
conf t
```

```
!
```

```
int f3/0
```

```
!
```

```
ip address 10.30.30.2 255.255.255.0
```

```
!
```

```
no sh
```

```
!
```

## Configuration iBGP Routeur 3

```
router bgp 65103
```

```
!
```

```
neighbor 192.168.2.2 remote-as 65101
```

```
!
```

```
network 192.168.2.0 mask 255.255.255.0
```

```
!
```

```
network 172.16.1.0 mask 255.255.255.0
```

```
!
```

## Configuration iBGP Routeur 4

```
no router bgp 65100
```

```
!
```

```
router bgp 65101
```

```
!
```

```
no neighbor 172.16.2.1 remote-as 65101
```

```
!
```

```
network 172.16.2.0 mask 255.255.255.0
```

```
!
```

```
network 10.10.10.0 mask 255.255.255.0
```

```
!
```

```
network 10.30.30.0 mask 255.255.255.0
```

```
!
```

```
end
```

```
!
```

```
conf t
```

```
!
```

```
int f2/0
```

```
!
```

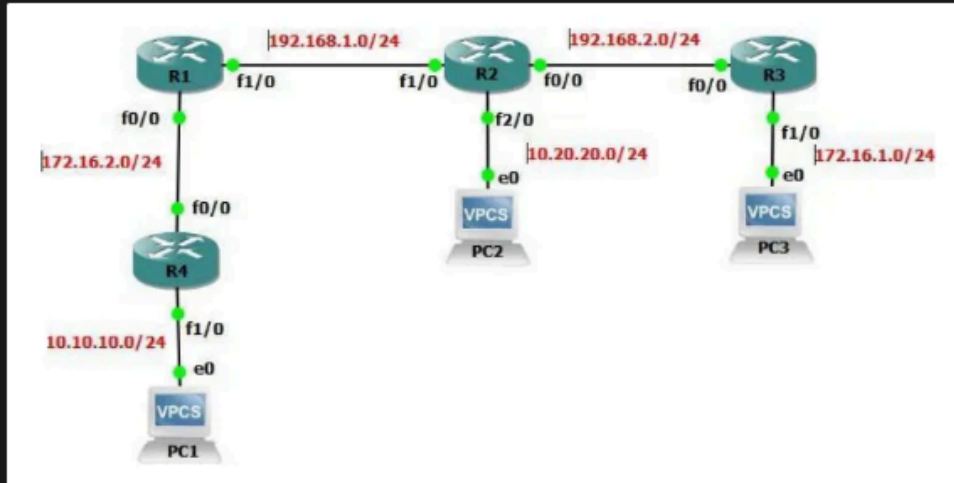
```
ip address 10.30.30.4 255.255.255.0
```

```
!
```

```
no sh
```

# TP1 - R302

On réalise ce schéma :



## Configuration IP :

On commence par mettre les adresses sur tout les équipements.

Configuration IP Routeur 1 :

```
en
!
conf t
!
interface FastEthernet0/0
ip address 172.16.2.1 255.255.255.0
no sh
duplex auto
speed auto
!
interface FastEthernet1/0
ip address 192.168.1.1 255.255.255.0
no sh
duplex auto
speed auto
!
```

#### Configuration IP Routeur 2 :



Plain Text



```
en
!
conf t
!
interface FastEthernet0/0
 ip address 192.168.2.2 255.255.255.0
 no sh
 duplex auto
 speed auto
!
interface FastEthernet1/0
 ip address 192.168.1.2 255.255.255.0
 no sh
 duplex auto
 speed auto
!
interface FastEthernet2/0
 ip address 10.20.20.2 255.255.255.0
 no sh
 duplex auto
 speed auto
!
```

#### Configuration IP Routeur 3 :

```
en
!
conf t
!
interface FastEthernet0/0
 ip address 192.168.2.3 255.255.255.0
 no sh
 duplex auto
 speed auto
!
interface FastEthernet1/0
 ip address 172.16.1.3 255.255.255.0
 no sh
 duplex auto
 speed auto
!
```

#### Configuration IP Routeur 4 :

```
en
!
conf t
!
interface FastEthernet0/0
 ip address 172.16.2.4 255.255.255.0
 no sh
 duplex auto
 speed auto
!
interface FastEthernet1/0
 ip address 10.10.10.4 255.255.255.0
 no sh
 duplex auto
 speed auto
!
```

Configuration IP PC1 :

```
ip 10.10.10.5/24 10.10.10.4
```

Configuration IP PC2 :

```
ip 10.20.20.5/24 10.20.20.2
```

Configuration IP PC3 :

```
ip 172.16.1.5/24 172.16.1.3
```

## Configuration eBGP :

On configure le eBGP sur tout les routeurs.

Configuration BGP Routeur 1 :

```
router bgp 65101
!
neighbor 192.168.1.2 remote-as 65102
!
neighbor 172.16.2.4 remote-as 65100
!
network 192.168.1.0 mask 255.255.255.0
!
network 172.16.2.0 mask 255.255.255.0
!
```

Configuration BGP Routeur 2 :

```
router bgp 65102
!
neighbor 192.168.1.1 remote-as 65101
!
neighbor 192.168.2.3 remote-as 65103
!
network 192.168.1.0 mask 255.255.255.0
!
network 192.168.2.0 mask 255.255.255.0
!
network 10.20.20.0 mask 255.255.255.0
!
```

Configuration BGP Routeur 3 :

```
router bgp 65103
!
neighbor 192.168.2.2 remote-as 65102
!
network 192.168.2.0 mask 255.255.255.0
!
network 172.16.1.0 mask 255.255.255.0
!
```

Configuration BGP Routeur 4 :

```
router bgp 65100
!
neighbor 172.16.2.1 remote-as 65101
!
network 172.16.2.0 mask 255.255.255.0
!
network 10.10.10.0 mask 255.255.255.0
!
```

Tout ping bien PC1 peut bien ping PC3 :

```
PC1> ping 172.16.1.5
84 bytes from 172.16.1.5 icmp_seq=1 ttl=60 time=121.611 ms
84 bytes from 172.16.1.5 icmp_seq=2 ttl=60 time=121.224 ms
84 bytes from 172.16.1.5 icmp_seq=3 ttl=60 time=120.784 ms
84 bytes from 172.16.1.5 icmp_seq=4 ttl=60 time=120.742 ms
84 bytes from 172.16.1.5 icmp_seq=5 ttl=60 time=121.637 ms
```

## Configuration iBGP :

On configure le iBGP sur R1, R2 et R4.

Configuration iBGP sur R1 :

```
router bgp 65101
!
neighbor 192.168.1.2 remote-as 65101
!
neighbor 172.16.2.4 remote-as 65101
!
network 192.168.1.0 mask 255.255.255.0
!
network 172.16.2.0 mask 255.255.255.0
!
```

Configuration iBGP sur R2 :

```
no router bgp 65102
!
router bgp 65101
!
neighbor 192.168.1.1 remote-as 65101
!
neighbor 192.168.2.3 remote-as 65103
!
neighbor 10.30.30.4 remote-as 65101
!
network 192.168.1.0 mask 255.255.255.0
!
network 192.168.2.0 mask 255.255.255.0
!
network 10.20.20.0 mask 255.255.255.0
!
network 10.30.30.0 mask 255.255.255.0
!
end
!
conf t
!
int f3/0
!
ip address 10.30.30.2 255.255.255.0
!
no sh
!
```



Configuration eBGP sur R3 :

```
router bgp 65103
!
neighbor 192.168.2.2 remote-as 65101
!
network 192.168.2.0 mask 255.255.255.0
!
network 172.16.1.0 mask 255.255.255.0
!
```

Configuration iBGP sur R4 :

```
no router bgp 65100
!
router bgp 65101
!
neighbor 172.16.2.1 remote-as 65101
!
neighbor 10.30.30.2 remote-as 65101
!
network 172.16.2.0 mask 255.255.255.0
!
network 10.10.10.0 mask 255.255.255.0
!
network 10.30.30.0 mask 255.255.255.0
!
end
!
conf t
!
int f2/0
!
ip address 10.30.30.4 255.255.255.0
!
no sh
```

## Bloquage réseau PC1 depuis R2 :

On veut empêcher R2 de trouver le réseau 10.10.10.0.

Configuration Routeur R4 :

```
route-map BLOCK-PC1 deny 10
!
match ip address 10
!
access-list 10 deny 10.10.10.0 0.0.0.255
!
router bgp 65101
!
neighbor 10.30.30.2 route-map BLOCK-PC1 out
```

```
!  
ip address 10.30.30.4 255.255.255.0  
!  
no sh
```

## Bloquage réseau PC1 depuis R2 :

On veut empêcher R2 de trouver le réseau 10.10.10.0.

Configuration Routeur R4 :

```
route-map BLOCK-PC1 deny 10  
!  
match ip address 10  
!  
access-list 10 deny 10.10.10.0 0.0.0.255  
!  
router bgp 65101  
!  
neighbor 10.30.30.2 route-map BLOCK-PC1 out
```

## Choix du Chemin : ATTENTION EN COURS

On configure R3 pour qu'il choisisse la route

Configuration R3 :

```
ip prefix-list NET_PC1 seq 5 permit 10.10.10.0/24  
!  
route-map PREF permit 10  
!  
match ip address prefix-list NET_PC1  
!  
set local-preference 200  
!  
exit  
!  
router bgp 65101  
!  
neighbor 10.30.30.2 route-map NET_PC1 out
```